

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI
COLLEGE OF ENGINEERING

B.Sc. (Engineering) Mid-Semester Examination, 2018
First Year

EE 172 ELECTRICAL MACHINES

MARCH, 2018

ONE HOUR

Index No.....

Year: one, two, three or four.

Each question is followed by four options lettered A to D. Find out the correct option for each question and circle without ambiguity the letter for the option you have chosen and also shade on the scannable sheet. Submit the question sheet together with the scannable.

1. Which of the following statements about dc compound generator are true?

- I It is cumulatively compounded.
- II Flat compounding is used to compensate for armature circuit voltage drop and also for the voltage drop in the feeder line between the generator and the load.
- III The terminal voltage increases with load if it is over-compounded.

- A I and II only
- B I and III only
- C II and III only
- D I, II and III

2. A long-shunt compound generator has a terminal voltage of 550 V when supplying a full load of 150 kW. The armature, shunt-field and series-field resistances are 0.04 Ω , 175 Ω and 0.25 Ω respectively. Calculate the emf generated in the armature in volts.

- A 568
- B 576
- C 584
- D 597

Use the information below to answer questions 3 and 4

A four-pole armature is wound with 564 conductors and driven at 800 rev/min, the flux per pole being 20 mWb. The current in each conductor is 60 A. The armature is lap-connected.

3. Calculate the value of the generated emf in volts.

- A 147
- B 151
- C 155
- D 163

4. Find the output current on full load in amps.

- A 210
- B 220
- C 230
- D 240

5. For a D. C. generator when the number of poles and the number of armature conductors is fixed, then which winding will give the higher e.m.f. ?

- A Lap winding
- B Wave winding
- C Either of (A) and (B) above
- D Depends on other features of design

6. Two generators A and B have 6-poles each. Generator A has wave wound armature while generator B has lap wound armature. The ratio of the induced e.m.f. in generator A and B will be
- A 2:3
 - B 3:1
 - C 3:2
 - D 1:3
7. The terminal voltage of a series generator is 150 V when the load current is 5 A. If the load current is increased to 10 A, the terminal voltage will be
- A 150 V
 - B less than 150 V
 - C greater than 150 V
 - D none of the above
8. The open circuit voltage of a compound generator is 250 V. At full load the terminal voltage
- A will be less than 250 V
 - B will always be 250 V
 - C may be greater or less than 250 V
 - D none of the above
9. In a D. C. generator the critical resistance can be increased by
- A increasing its field resistance
 - B decreasing its field resistance
 - C increasing its speed
 - D decreasing its speed
10. The wave-connected armature of a four-pole dc generator is required to generate an emf of 520 V when driven at 660 rev/min. Calculate the flux per pole required if the armature has

144 slots with 2 coil sides per slot, each consisting of 3 turns.

- A 20.4 mWb
- B 27.3 mWb
- C 30.2 mWb
- D 25 mWb

11. A four-pole motor is fed at 440V and takes an armature current of 50 A. The resistance of the armature circuit is 0.28 Ω . The armature winding is wave-connected with 888 conductors and the useful flux per pole is 0.023 Wb. Calculate the speed.
- A 620 rev/min
 - B 500 rev/min
 - C 626 rev/min
 - D 690 rev/min

Use the information below to answer questions 12 and 13

A dc machine is connected as a shunt generator. When operated at rated speed of 1500 rev/min, the O.C. curve is approximated

$$\text{by } E = \frac{367I_f}{0.233 + I_f}.$$

The armature circuit resistance is 0.14 Ohm. The shunt field resistance is adjusted to be 459 ohms

12. Find the no load terminal voltage.
- A 200 V
 - B 180 V
 - C 130 V
 - D 260 V
13. Find the field current
- A 0.43 A
 - B 0.39 A
 - C 0.57 A
 - D 0.28 A

Use the information below to answer questions 14 and 15

It is desired to level-compound a shunt generator so that the terminal voltage at full load of 100A is equal to that at no load. When connected as a shunt generator the field current had to be increased from 1.76A to 4.3A in order to keep the terminal voltage the same at no load and full load. If the machine is to be connected long-shunt and the turns per pole on the shunt field is 1100,

14. Calculate the number of series turns/pole required
 - A 7
 - B 28
 - C 14
 - D 20
15. Calculate the generated emf that gives a terminal voltage of 220 V given that the shunt field and armature circuit resistances are $125\ \Omega$ and $0.12\ \Omega$ respectively.
 - A 240.6 V
 - B 232.2 V
 - C 200.8 V
 - D 222.1 V
16. If B is the flux density, l the length of conductor and v the velocity of conductor, then induced e.m.f. is given by
 - A Blv
 - B Blv^2
 - C Bl^2v^2
 - D Bl^2v
17. The function of pole shoes in the case of D.C. machine is
 - A to reduce the reluctance of the magnetic path
 - B to spread out the flux to achieve uniform flux density

- C to support the field coil
 - D to discharge all the above functions
18. The number of brushes in a commutator depends on
 - A speed of armature
 - B type of winding
 - C voltage
 - D amount of current to be collected
19. In D. C. generators on no-load, the air gap flux distribution in space is
 - A sinusoidal
 - B triangular
 - C pulsating
 - D flat topped
20. Which of the following D. C. generators will have negligible terminal voltage while running on no-load
 - A series generator
 - B shunt generator
 - C compound generator
 - D separately excited generator

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